SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 1 – Data Interpretation
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	CSE - AI	**Date:**	24/08/2026

Code of Conduct

I N. Tharun Sai (192472382) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: N. Tharun Sai

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

Annexure A – Data Interpretation

1. Frame Semantics in Banking Customer-Service Systems

A digital banking platform uses semantic frames to understand customer requests. The NLP system identifies the **action, agent, object, and destination** involved in each request.

Customer Query	Generated Semantic Frame
"Transfer ₹20,000 from my savings account to my son's account."	TRANSFER(SourceAccount, Amount, DestinationAccount, Customer)
"Block my debit card immediately."	BLOCK(DebitCard, Customer)
"Send my transaction statement to my email."	SEND(TransactionStatement, Email, Customer)
"Increase my daily withdrawal limit."	MODIFY(WithdrawalLimit, Customer)

Additional transaction logs show:

Query ID	Actual User Intent	System Interpretation
B1	Transfer money to another account	Transfer money
B2	Block debit card	Block debit card
B3	Receive transaction statement by email	Send statement to email
B4	Increase withdrawal limit	Decrease withdrawal limit

Tasks:

1. Analyze the semantic frame of each banking request and identify the relationship between the action and its participants.
2. Identify the query in which the semantic interpretation is incorrect and explain why.
3. Analyze how incorrect identification of semantic roles can lead to an incorrect banking operation.
4. Recommend a frame-based semantic analysis approach to improve the reliability of banking conversational systems.

2. First-Order Predicate Calculus for Smart Agriculture

A smart agricultural system monitors crop fields using sensors and logical rules.

Field Data:

Field	Soil Condition	Irrigation Status	Crop
F1	Dry	Available	Rice
F2	Wet	Available	Wheat
F3	Dry	Unavailable	Maize
F4	Wet	Unavailable	Rice

The system uses the following predicate rules:

- Dry(x) ∧ IrrigationAvailable(x) → NeedsWater(x)
- NeedsWater(x) → Irrigate(x)
- Wet(x) → ¬NeedsWater(x)
- IrrigationUnavailable(x) → ¬Irrigate(x)
- Rice(x) ∧ NeedsWater(x) → PriorityIrrigation(x)

Tasks:

1. Represent the field observations using First-Order Predicate Calculus.
2. Using the given rules, determine which fields require irrigation.
3. Determine whether F1 and F3 should be treated differently by the automated irrigation controller.
4. Analyze whether predicate logic alone is sufficient for agricultural decision support when sensor information is incomplete or uncertain.

3. Word Sense Disambiguation in a Travel Recommendation System

A travel platform receives search queries containing ambiguous words. The recommendation engine uses surrounding words and user interactions to determine the intended meaning.

Search Query	Possible Sense 1	Possible Sense 2
"Amazon tour"	Amazon rainforest	Amazon company
"Safari booking"	Wildlife tour	Web browser
"Java vacation"	Java island	Java programming language
"Cruise package"	Ship journey	Cruise software/project

The system records the following user interactions:

Query	User Interaction
Amazon tour	Viewed rainforest trekking packages
Safari booking	Selected wildlife resort packages

Query	User Interaction
Java vacation	Viewed hotels in Bali and Java
Cruise package	Viewed Mediterranean ship packages

However, another user searches: **"Safari download for Windows"** and clicks a browser download page.

Tasks:

1. Determine the intended sense of the ambiguous terms using the available context.
2. Identify the lexical and contextual cues that distinguish the possible senses.
3. Analyze why the same word may require different interpretations for different users.
4. Design an industrial-scale WSD strategy using query context, user history, embeddings, and click behaviour to improve travel recommendations.

4. Syntax-Driven Semantic Analysis in Legal Document Processing

A legal NLP system analyzes sentences from contracts and assigns semantic roles based on their syntactic structures.

The system produces the following analyses:

Legal Sentence	Parse Structure	Generated Semantic Interpretation
"The supplier delivered the equipment to the buyer."	Subject–Verb–Object	Supplier = Agent, Equipment = Object, Buyer = Recipient
"The buyer received the equipment from the supplier."	Subject–Verb–Object	Buyer = Agent, Equipment = Object, Supplier = Recipient
"The company terminated the contract after the violation."	Subject–Verb–Object	Company = Agent, Contract = Object
"The contract was terminated by the company."	Passive Construction	Contract = Agent, Company = Object

Tasks:

1. Analyze how syntactic structure, including active and passive constructions, influences semantic-role assignment.
2. Identify the incorrect semantic-role assignments in the given interpretations.
3. Explain how confusing grammatical subject with semantic agent can affect automated legal information extraction.
4. Recommend a syntax-driven semantic analysis architecture that can correctly handle active voice, passive voice, prepositional phrases, and long-distance dependencies in legal documents.

1. Frame Semantics in Banking Customer-Service Systems

1. Semantic Frame Analysis

A **semantic frame** represents an action/event and the participants involved in that action.

Customer Query	Frame/Action	Semantic Roles
“Transfer ₹20,000 from my savings account to my son's account.”	**TRANSFER**	Customer = Agent, ₹20,000 = Amount, Savings Account = Source, Son's Account = Destination
“Block my debit card immediately.”	**BLOCK**	Customer = Agent, Debit Card = Object
“Send my transaction statement to my email.”	**SEND**	Customer = Agent, Transaction Statement = Object, Email = Destination
“Increase my daily withdrawal limit.”	**MODIFY**	Customer = Agent, Withdrawal Limit = Object, Increase = Direction

Frame Representations

TRANSFER(Customer, SavingsAccount, ₹20,000, SonAccount)

BLOCK(Customer, DebitCard)

SEND(Customer, TransactionStatement, Email)

MODIFY(Customer, WithdrawalLimit, Increase)

2. Incorrect Semantic Interpretation

The incorrect interpretation is B4.

Query ID	Actual Intent	System Interpretation	Result
B1	Transfer money to another account	Transfer money	Partially correct
B2	Block debit card	Block debit card	Correct
B3	Receive statement by email	Send statement to email	Correct
B4	Increase withdrawal limit	Decrease withdrawal limit	Incorrect

The system identified the correct object, WithdrawalLimit, but incorrectly identified the direction of modification.

Correct:

MODIFY(WithdrawalLimit, Increase)

Incorrect:

MODIFY(WithdrawalLimit, Decrease)

3. Effect of Incorrect Semantic Roles

Incorrect semantic-role identification can result in the wrong banking operation.

Example:

User: "Increase my withdrawal limit."

Correct:

WithdrawalLimit → Increase

Incorrect:

WithdrawalLimit → Decrease

This may cause:

- Incorrect account modification
- Transaction failures
- Customer complaints
- Financial risk

4. Recommended Frame-Based Approach

A reliable banking NLP system can use the following pipeline:
Customer Query
↓
Tokenization
↓
Intent Detection
↓
Frame Identification
↓
Semantic Role Labelling
↓
Entity Extraction
↓
Role Validation
↓
Banking Operation
The system should specifically validate:

- Action
- Agent/Customer
- Amount
- Source
- Destination
- Object
- Modification direction

This improves the reliability and safety of conversational banking systems.

2. First-Order Predicate Calculus for Smart Agriculture

1. Field Observations in FOPC

The field information can be represented using predicates.
F1
Dry(F1)
IrrigationAvailable(F1)
Rice(F1)
F2
Wet(F2)
IrrigationAvailable(F2)
Wheat(F2)
F3
Dry(F3)
IrrigationUnavailable(F3)
Maize(F3)
F4
Wet(F4)
IrrigationUnavailable(F4)
Rice(F4)

2. Given Predicate Rules

Dry(x) ∧ IrrigationAvailable(x) → NeedsWater(x)

NeedsWater(x) → Irrigate(x)
Wet(x) → ¬NeedsWater(x)
IrrigationUnavailable(x) → ¬Irrigate(x)
Rice(x) ∧ NeedsWater(x) → PriorityIrrigation(x)

3. Determine Which Fields Require Irrigation

F1
Dry(F1)
IrrigationAvailable(F1)
Therefore:
NeedsWater(F1)
Irrigate(F1)
Since F1 contains rice:
PriorityIrrigation(F1)
Decision: F1 should be irrigated.
F2
Wet(F2)
Therefore:
¬NeedsWater(F2)
Decision: F2 does not require irrigation.
F3
Dry(F3)
IrrigationUnavailable(F3)
F3 is dry, but irrigation is unavailable.
Therefore:
¬Irrigate(F3)
Decision: F3 cannot be irrigated.
F4
Wet(F4)
IrrigationUnavailable(F4)
Therefore:
¬NeedsWater(F4)
¬Irrigate(F4)
Decision: F4 does not require irrigation.
Final Result

Field	Soil	Irrigation Availability	Decision
F1	Dry	Available	**Irrigate**
F2	Wet	Available	**Do not irrigate**
F3	Dry	Unavailable	**Cannot irrigate**
F4	Wet	Unavailable	**Do not irrigate**

4. F1 vs F3

F1 and F3 should be treated differently.
Both fields are **dry**, but:
F1 → Irrigation available → Irrigate
F3 → Irrigation unavailable → Do not irrigate
Also, F1 grows rice, so it receives:

PriorityIrrigation(F1)

5. Is Predicate Logic Sufficient?

No.

Predicate logic generally represents information as true or false. Real agricultural systems often have uncertain or incomplete sensor information.

For example:

Soil moisture = 40%

Sensor confidence = 70%

The system may not be completely certain whether the soil is dry.

Other problems include:

- Sensor failure
- Missing data
- Conflicting sensor readings
- Weather uncertainty
- Changing soil conditions

Therefore, predicate logic can be combined with fuzzy logic, probability, sensor confidence, and machine learning for better agricultural decision-making.

3. Word Sense Disambiguation in a Travel Recommendation System

1. Intended Sense of Each Query

Search Query	Intended Sense	Evidence
Amazon tour	Amazon rainforest	User viewed rainforest trekking packages
Safari booking	Wildlife tour	User selected wildlife resort packages
Java vacation	Java Island	User viewed hotels in Bali and Java
Cruise package	Ship journey	User viewed Mediterranean ship packages
Safari download for Windows	Safari web browser	User clicked a browser download page

2. Lexical and Contextual Cues

Amazon

Amazon + tour

Amazon + rainforest

Amazon + trekking

These words indicate the Amazon rainforest.

Safari

Safari + booking

Safari + wildlife

Safari + resort

indicates a wildlife tour.

But:

Safari + download

Safari + Windows

Safari + browser

indicates the Safari web browser.

Java

Java + vacation

Java + hotels

Java + Bali

indicates Java Island.

Whereas:

Java + programming
Java + code
Java + compiler
would indicate the Java programming language.
Cruise
Cruise + package
Cruise + Mediterranean
Cruise + ship
indicates a ship journey.

3. Why the Same Word Has Different Meanings

A word can have multiple meanings depending on:

- Surrounding words
- Search query
- User history
- Previous clicks
- User interests
- Domain/context

For example:
Safari booking
↓
Wildlife tourism
but:
Safari download for Windows
↓
Web browser
Therefore, WSD cannot depend only on the ambiguous word itself.

4. Industrial-Scale WSD Strategy

A travel recommendation system can use:
Search Query
↓
Tokenization
↓
Context Analysis
↓
Candidate Sense Generation
↓
Contextual Embeddings
↓
User History
↓
Click Behaviour
↓
Sense Ranking
↓
Travel Recommendation

4. Syntax-Driven Semantic Analysis in Legal Document Processing

1. Syntax and Semantic Roles

Syntactic structure helps identify semantic roles, but the grammatical subject is not always the semantic Agent.

Sentence 1

“The supplier delivered the equipment to the buyer.”

Entity	Semantic Role
Supplier	Agent
Equipment	Object/Theme
Buyer	Recipient

The given interpretation is **correct**.

Sentence 2

“The buyer received the equipment from the supplier.”

Correct interpretation:

Entity	Semantic Role
Buyer	Recipient
Equipment	Object/Theme
Supplier	Agent/Source

The given interpretation says:
Buyer = Agent
Supplier = Recipient
This is **incorrect**.

Sentence 3

“The company terminated the contract after the violation.”

Correct interpretation:

Entity	Semantic Role
Company	Agent
Contract	Object/Patient
Violation	Context/Cause

The given interpretation of company and contract is **correct**, although the role of the violation is not explicitly represented.

Sentence 4

“The contract was terminated by the company.”

Correct interpretation:

Entity	Semantic Role
Contract	Patient/Object
Company	Agent

The given interpretation:
Contract = Agent
Company = Object
is **incorrect**.

2. Incorrect Semantic-Role Assignments

Sentence	Incorrect Assignment	Correct Assignment
Buyer received equipment from supplier	Buyer = Agent	Buyer = Recipient
	Supplier = Recipient	Supplier = Agent/Source
Contract was terminated by company	Contract = Agent	Contract = Patient/Object
	Company = Object	Company = Agent

3. Why Subject ≠ Agent

Consider:
The contract was terminated by the company.
The grammatical subject is:
The contract
But the entity performing the action is:
The company
Therefore:
Grammatical Subject ≠ Always Semantic Agent
If a legal NLP system assumes that the subject is always the Agent, it could incorrectly conclude:
Contract → terminated the company
instead of:
Company → terminated the contract
This can seriously affect:

- Liability extraction
- Contract analysis
- Legal event extraction
- Obligation identification
- Dispute analysis

4. Recommended Syntax-Driven Architecture

A robust legal NLP system should use:
Legal Document
↓
Sentence Segmentation
↓
Tokenization
↓
POS Tagging
↓
Dependency Parsing
↓
Active/Passive Voice Detection
↓
Semantic Role Labelling
↓
Coreference Resolution
↓
Legal Entity & Event Extraction
↓
Structured Legal Knowledge

Active Voice
Supplier → delivered → Equipment
Therefore:
Supplier = Agent
Equipment = Object
Passive Voice
Contract ← terminated ← Company
The system must recognize the passive construction and assign:
Company = Agent
Contract = Patient
Prepositional Phrases
The system should analyze phrases such as:
to the buyer
from the supplier
after the violation
because they provide important semantic information such as:

- Recipient
- Source
- Time
- Cause
- Location

Long-Distance Dependencies
Legal sentences are often lengthy and contain multiple clauses. Dependency parsing and semantic-role labeling help connect the correct entities even when they are separated by many words.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1	3	3	4	4	3	17	83
2	4	4	3	3	3	17	
3	3	3	4	4	3	17	
4	3	3	4	4	4	18	

Faculty In-charge	Course Coordinator	Program Director
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Date: ____________________	Date: ____________________	Date: ____________________